

FlavorExplorers - Milestone 1

Imane Benkamoun, Quirin Bitter, Sofia Taouhid

March 2026

1 Dataset

Our project integrates primary and secondary datasets to examine how recipe nutritional profiles relate to emotional appeal and purchasing patterns:

- **Kaggle Recipes:** Contains ~62,000 diverse everyday dishes [10]. We enriched this data using an LLM to assign a binary “healthiness” label based on ingredient composition.
- **Flavor Network:** A bipartite dataset containing 1,530 food ingredients linked to 1,106 flavor compounds via 36,781 edges [4]. This allows us to calculate the “flavor complexity” of any dish.
- **FooDB:** We extract macro- and micronutrient profiles, isolating specific neuro-active compounds like L-Tryptophan and Magnesium [9].
- **Secondary Datasets:** We engineered a custom `mood_mapping.csv` to translate FooDB’s chemical data into categorical mood scores. We also use UK household expenditures, tracking weekly per-person food spending from 1974 to 2024 across 328 categories [8, 7].

Preprocessing We harmonized these sources by bridging the Flavor Network and FooDB. Aligning row indices, standardizing naming conventions, and applying a fuzzy matching algorithm we successfully linked 315 core ingredients. The Kaggle dataset requires parsing raw ingredient strings, while the structured UK expenditure data mainly requires filtering to prevent double-counting via root categories.

Strengths & Limitations The primary strengths of our data are its massive scale and multi-modality. However, ingredient processing remains inherently noisy, and the biological links between meals, emotions, and healthiness are partially subjective. Thus, our project serves as a data-driven visualization of food’s position between pleasure and health, rather than a strict diagnostic tool.

2 Problematic

The Comfort Food Paradox: Does pleasure come at the expense of health? This project investigates a deeply familiar intuition about food: does the pleasure of eating inherently come at the expense of our physical health? In everyday life, people often oppose tasty, comforting food and healthy food, as if nutritional value must necessarily cost us culinary enjoyment. Rich meals are commonly associated with immediate satisfaction, while healthier dishes are frequently perceived as emotionally unrewarding. We want to determine whether the happiness we derive from comfort food is a true biological reality driven by mood-boosting nutrients, or merely a sensorial illusion triggered by flavor additives and sugar.

Hypothesis & Methodology We hypothesize that less healthy recipes emphasize pleasure-related signals far more than healthier ones. To visualize this at scale, we operationalize “pleasure” as a mathematical combination of a dish’s flavor complexity and its density of emotion-associated chemical compounds. In parallel, we estimate “healthiness” using ingredient-based LLM labels. This methodology yields two comparable dimensions across thousands of dishes: a physical health axis and a neurological pleasure axis.

Objectives & Target Audience Our main objective is to visualize whether these two axes are strictly opposed in practice. Are unhealthy dishes consistently associated with ingredients linked to stronger pleasure effects, and do healthy recipes cluster in lower-pleasure regions? Secondly, we will map these findings onto UK expenditure data to examine which food-related emotions drive consumer spending. Our motivation is to help users identify the culinary “Goldilocks Zone” dishes scoring high on flavor complexity and biological mood stabilization without failing standard health metrics. Our target audience includes dietitians, menu planners, and health-conscious consumers.

3 Exploratory Data Analysis

3.1 Kaggle Recipes Dataset Exploration

Our exploratory analysis validates our preprocessing and reveals structural insights. Plotting the top food categories demonstrates a clear divide regarding healthiness: main dishes are relatively healthy, whereas categories like cakes and cookies skew heavily unhealthy (figure 2). Ingredient frequency strongly reinforces this (figure 3). Healthy recipes overwhelmingly feature ingredients like garlic and olive oil, while unhealthy recipes are dominated by baking staples like white sugar and butter. Isolating ingredients disproportionately associated with one specific class confirms our healthiness classification matches established nutritional intuition (figure 4).

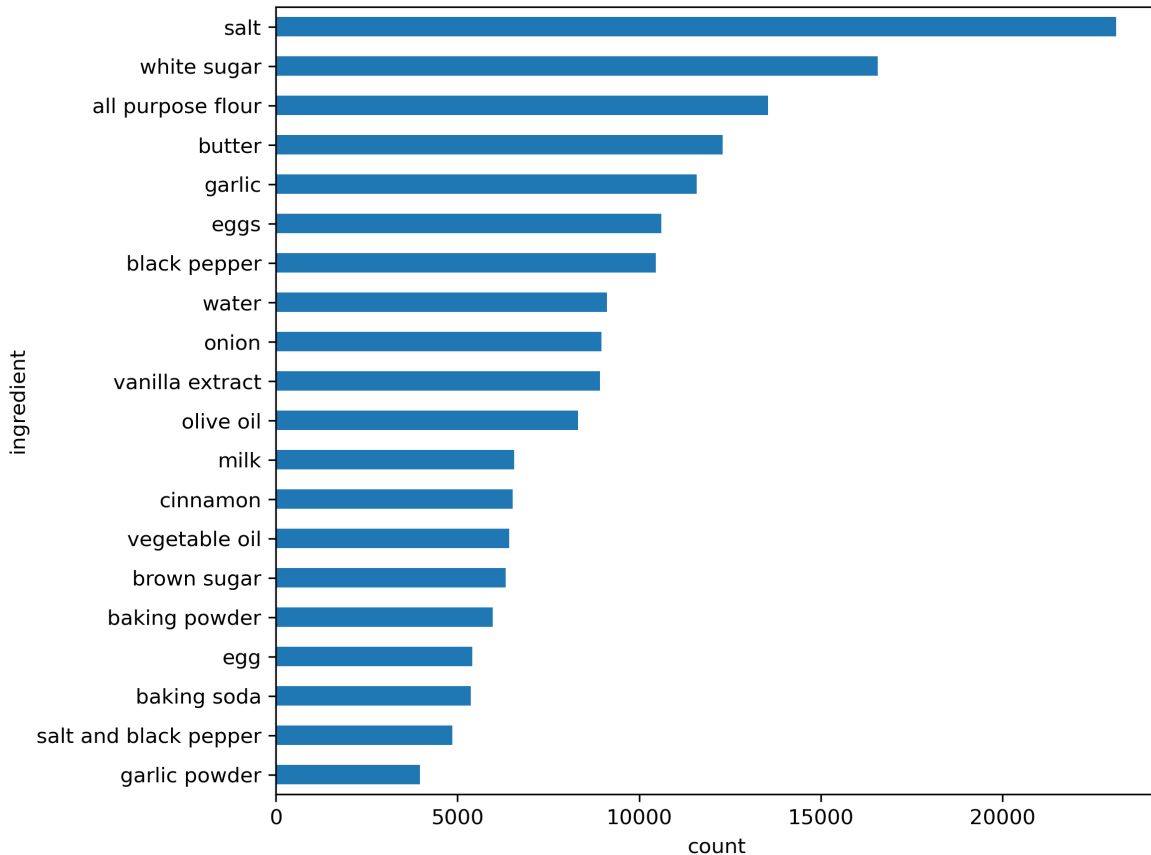


Figure 1: The 20 most common ingredients over all 62 126 recipes.

3.2 Flavor Network & FooDB Exploration

First, we calculated the “Flavor Complexity” of individual ingredients. The resulting distribution plot reveals a massive right-skewed, long-tail distribution (figure 5). Most ingredients share very few chemical compounds, but a select few act as massive hubs. Interestingly, the most complex ingredients are highly skewed toward beverages (Black Tea: 239 compounds, Coffee: 234) and meats (Grilled Beef: 217). To test the technical feasibility of our pipeline, we ran an advanced string-matching script. Standardizing strings and applying a `difflib` fuzzy matching algorithm¹ successfully bridged 315 core ingredients between the two databases, yielding a 20.57% match rate. Grouping these matched ingredients by their FooDB categories reveals a balanced baseline dominated by fruits, herbs, and vegetables (figure 6).

¹We used Python’s built-in `difflib` module with an 80% similarity score.

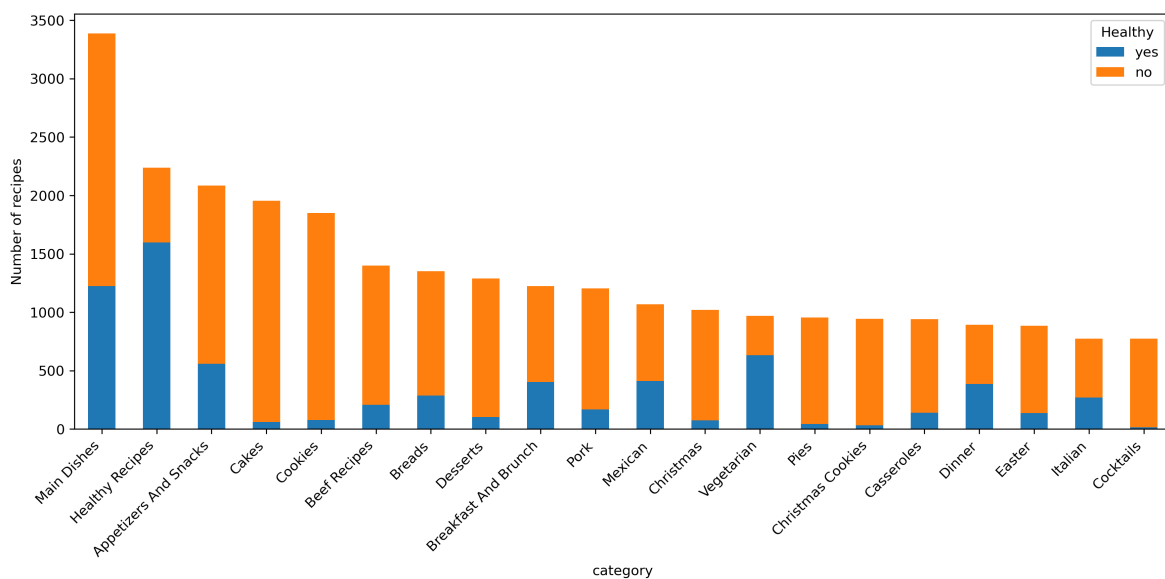


Figure 2: Health Label Distribution Across Top Categories of the Kaggle Recipes Dataset.

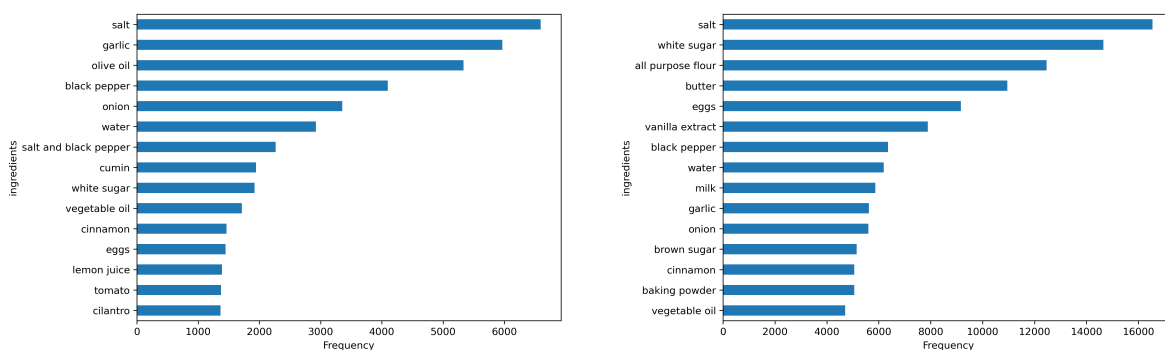


Figure 3: Most frequent ingredients in healthy recipes (left, 18 600 recipes) vs. unhealthy recipes (right, 43 526 recipes).

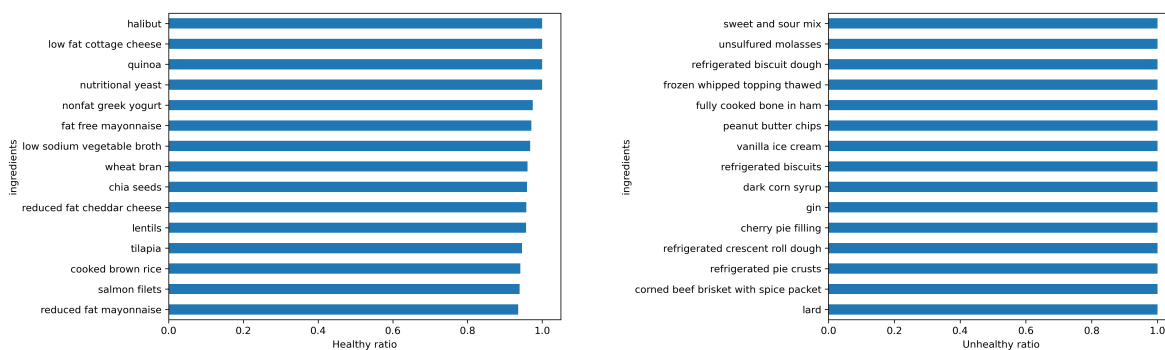


Figure 4: Ingredient association ratio with healthy recipes (left) vs. unhealthy recipes (right).

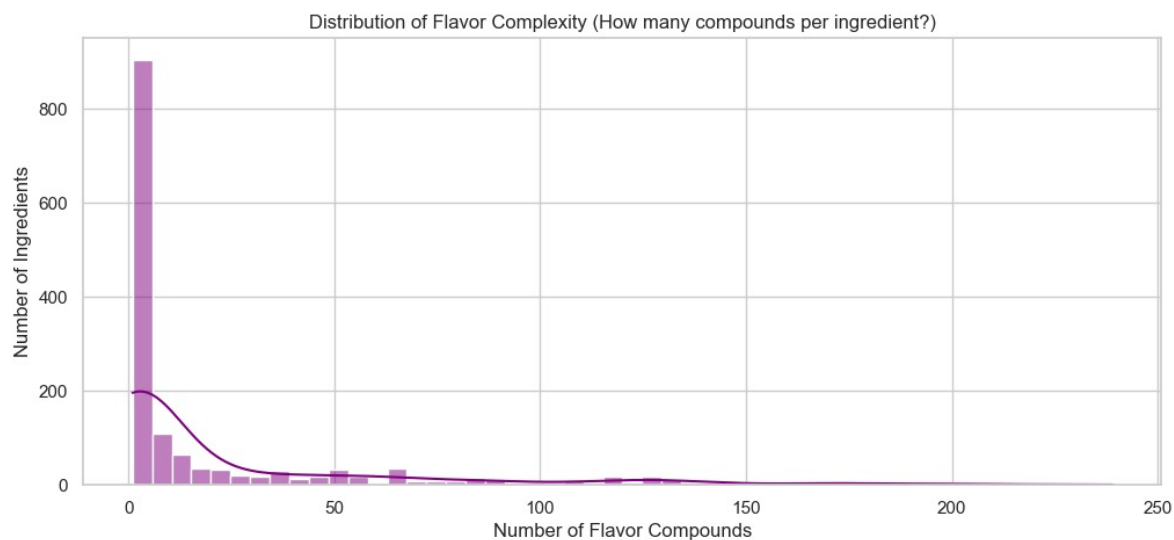


Figure 5: Flavor Complexity Distribution: How many compounds does an ingredient have?

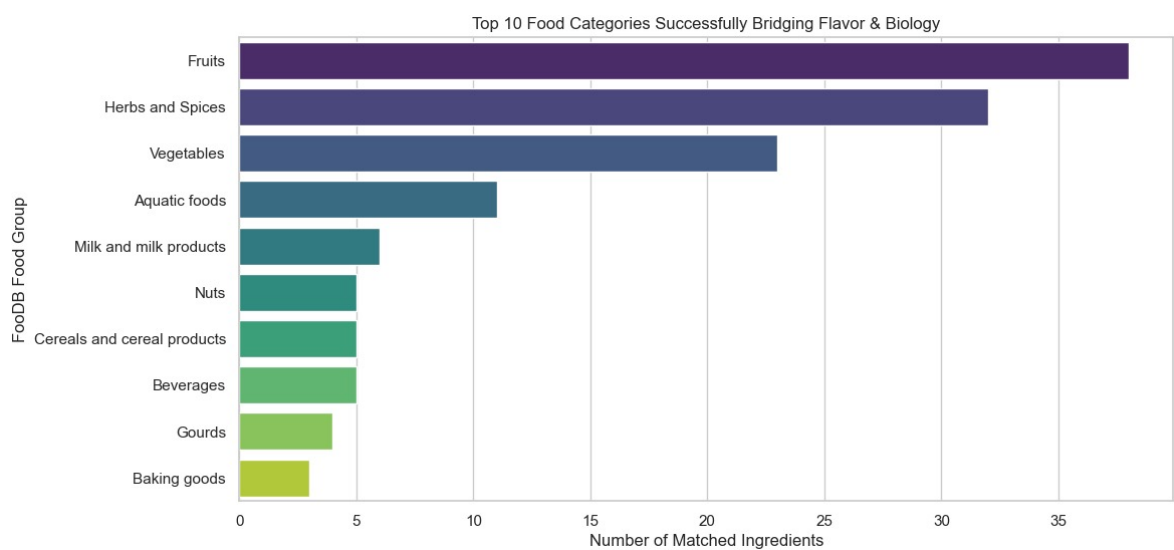


Figure 6: Ingredients successfully matched to flavor compounds, categorized by their FoodDB group.

3.3 UK Expenditure Dataset

This dataset provides 14,743 observations of weekly per-person food spending from 1974 to 2024². With 328 tracked categories, the matrix coverage sits at an excellent 93%, with a low Relative Standard Error averaging 5–10%, providing a reliable temporal foundation for mapping spending habits to our nutritional findings.

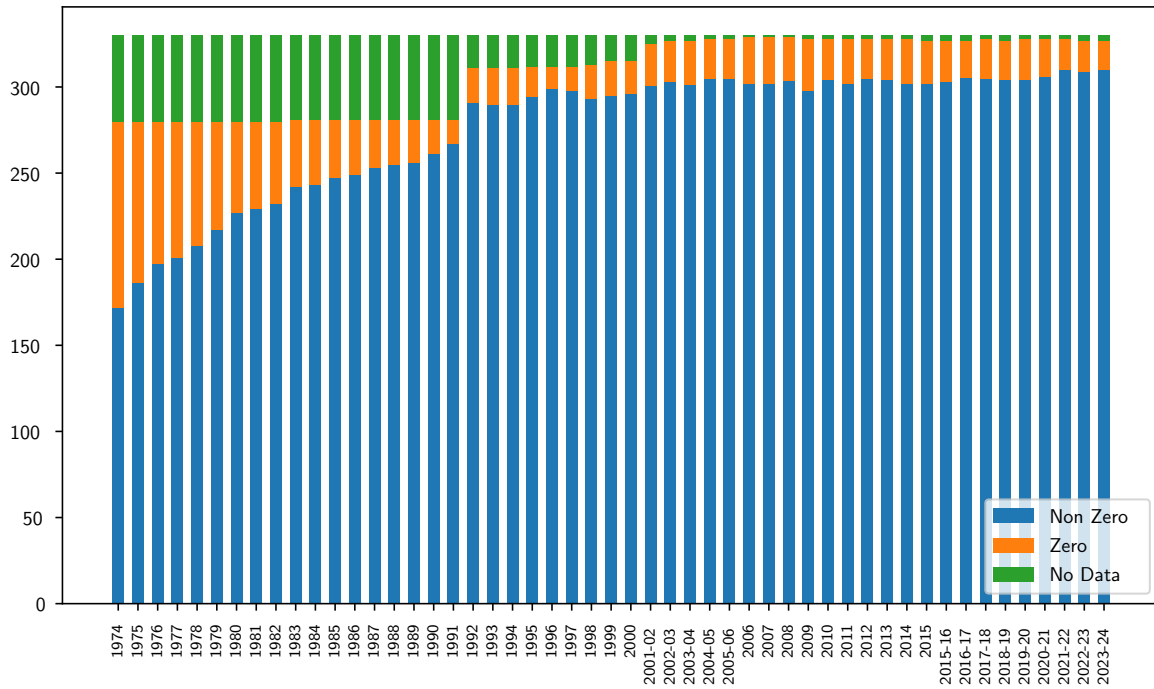


Figure 7: Bar plot showing the UK expenditure dataset: for each year, counts of categories with data points, categories with zero values, and categories with missing values.

²Please note that the data for the years 2001–2005 and 2016–2024 are provided with overlapping periods (e.g., 2001–2002). A note in the dataset explains that this change occurred because different figures were used starting from 2001 onward. However, it is unclear why the data is provided annually for the years 2006–2015.

4 Related Work

The core of our sensorial data relies on the 2011 *Nature Scientific Reports* paper by Ahn et al., “Flavor network and the principles of food pairing” [1]. Ahn’s work famously visualized how different global cuisines either embrace or reject flavor-compound overlap.

Other work covered gender differences in dietary habits for salty and umami tastes [3] or oral sensation from the consumption of alcoholic beverages [5].

Furthermore, is it well known that consumer behavior is influenced and likely linked to emotional events [2, 6]. Further is it known, that the type of expenditure influences people’s overall happiness [11]. However, no work known to us, links different food expenditures to happiness. We hope that, pairing our insights from “What tastes good and keeps us healthy” with our expenditure dataset, may allow statements regarding taste and health related spendings.

Originality Previous approaches focused almost exclusively on the taste aspect of food networks or purely on the nutritional aspect. Our approach is highly original because it links these isolated domains in our *engineered datasets*. By injecting a biological/health layer into the flavor network, we shift the narrative from “What tastes good together?” to “What tastes good, keeps us healthy, and biochemically makes us feel good?”.

Pairing our findings with the expenditure data set may secondarily allow us to figure out: for which feeling do people spend their money?

Inspiration Visually, we draw inspiration from interactive data journalism pieces found on *The Pudding*³ and *FiveThirtyEight*⁴.

³<https://pudding.cool/>

⁴<https://fivethirtyeight.com/tag/data-visualization/#content>

References

- [1] Yong-Yeol Ahn, Sebastian E Ahnert, James P Bagrow, and Albert-László Barabási. Flavor network and the principles of food pairing. *Scientific reports*, 1(1):196, 2011.
- [2] Qingyin Cai and Qingxiao Li. Emotional shocks and consumer spending. *Journal of Economic Behavior Organization*, 241:107357, 2026.
- [3] Camilla Cattaneo, Sara Spinelli, Caterina Dinnella, Cristina Proserpio, Erminio Monteleone, Ella Pagliarini, and Monica Laureati. Patterns of sensory and hedonic responses for salty and umami tastes and their impact on food familiarity, consumption, and nutritional status: A gender-based analysis from a large population sample. *Current Research in Food Science*, 10:100970, 2025.
- [4] Ling Cheng. lingcheng99/flavor-network. <https://github.com/lingcheng99/Flavor-Network/tree/master>. Accessed: 2026-03-16.
- [5] Maria Carla Cravero, Monica Laureati, Sara Spinelli, Federica Bonello, Erminio Monteleone, Cristina Proserpio, Maria Rosa Lottero, Ella Pagliarini, and Caterina Dinnella. Profiling individual differences in alcoholic beverage preference and consumption: New insights from a large-scale study. *Foods*, 9(8), 2020.
- [6] Sardar Abbas Haider Khan. Stress and spending dynamics: An economically framed study of emotional expenditures in youth. *Review of Applied Management and Social Sciences*, 8(4):1479–1492, 2025.
- [7] N/A. Population estimates time series dataset. <https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/datasets/populationestimatestimeseriesdataset>. Accessed: 2026-03-16.
- [8] N/A. Uk - expenditure. <https://www.gov.uk/government/statistical-data-sets/family-food-datasets>. Accessed: 2026-03-16.
- [9] N/A. www.foodb.ca. <https://foodb.ca/downloads>. Accessed: 2026-03-16.
- [10] Prashant Singh. Recipes dataset: 64k dishes. <https://www.kaggle.com/datasets/prashantsingh001/recipes-dataset-64k-dishes>. Accessed: 2026-03-16.
- [11] Säde Stenlund, Yingchi Guo, Jason Rights, Ryan Dwyer, and Elizabeth Dunn. How spending decisions shape happiness in everyday life. *Communications Psychology*, 2(1):124, 2024.